RECONCILIATION AGENT · SOURCE SIMULATOR

How the data fits together

Where each record comes from, how one item is followed from the production plan to the warehouse and into Oracle, and why the agent compares totals per item rather than order line against order line.

For **warehouse and operations leaders**, Lehigh Valley bakery (UBD)
November 2026 · application version 0.2

1 The short version
2 The four systems
3 Where the data comes from
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All examples use simulated data. Item numbers run from 900101 to 900112, and every product and customer name starts with "SIM". Where this guide describes Bimbo's own reports, it gives only counts and formats, never real items, quantities or customers. Nothing in the simulator changes a real MIPS, WMS or Oracle system.

1 · THE SHORT VERSION

Five things to know

The bakery runs on four systems that each describe the same bread from their own angle. The agent compares them so that differences are found in minutes rather than by comparing screens by hand.

4	8	5 + 3	19
systems: MIPS, ODI, WMS, Oracle	record types in the simulator	5 from Bimbo reports, 3 added by us	checks run on every change

1 Each system answers a different question

MIPS plans what to make. ODI records what the plant made. WMS receives, allocates and ships it to customers. ORACLE books the stock and the money. None of them holds the whole picture.

2 No order-line number is shared by all the systems

MIPS splits an item's demand into plan lines (customer orders and forecast) and does not record customers. WMS splits the same demand by customer and has no MIPS line numbers. One line cannot be matched directly to the other, because the systems split the demand in different ways.

3 The item number is the common thread

The agent gathers **every line for the same item and sales day** in each system, adds them up, and compares the totals. Item numbers are cleaned first, because each system writes them differently. Descriptions are never used for matching, because they differ between systems.

4 Some records do link directly, and the agent follows those links

A production run is linked to its receipt ticket by the run code. The ticket is linked to its warehouse transaction by the ticket number, and the transaction to its Oracle posting by the transaction number. On the customer side, item lines are linked to their order by the customer name, and shipments to the order by the order number. Where a chain breaks, the agent raises a flag.

5 Systems do not update each other, so gaps open up

A change in one system stays in that system until a person, an interface or an approved correction changes the others. These differences are what the agent finds and what the worklist shows. The simulator lets us create these differences on purpose and watch the agent find them.

What this means for the warehouse. When the worklist flags an item, the evidence is the set of records behind it in every system: the plan lines, the runs, the customer lines, the receipts and the postings. Section 6 shows this for one item, step by step.

Who knows what

The same loaf passes through four systems on its way from the plan to the customer. Each system records only its own part.

Plan
How many of each item for the sales day

›

Mix and bake
Dough batches, production runs, units wrapped

›

Receive
Product arrives from the plant on a ticket

›

Allocate and ship
Customer orders are picked, loaded and shipped

›

Book
Every stock movement is posted to inventory and finance

MIPS

Planning

"What should we make, for which sales day?"

Holds	Plan lines: customer sales orders (from BUSS) and forecast lines (from OCloud supply planning)
Calls the item	**FG Oracle Ref**, plus the old **FG BUSS Code**
Does not know	Which customer a line is for, or what was actually made
Simulator tab	Order lines

ODI

Production records

"What did the plant mix, bake and wrap?"

Holds	One record per item per production run, and one mixing record per run
Calls the item	**OracleFGNumber**, plus **LegacyFGNumber** (the same value as the BUSS code)
Does not know	Customers, or what happens after wrapping
Simulator tabs	Finished goods, Batches

WMS

Warehouse

"What came in, who gets it, and what left on which trailer?"

Holds	Customer order lines by item, order headers, receipt tickets and stock movements
Calls the item	**Item Nbr**. In the real report this is stored as a number, not text
Does not know	The MIPS plan lines, or how stock was netted in planning
Simulator tabs	Item lines, Orders, Tickets, Transactions

ORACLE

Oracle Fusion: inventory and finance

"What stock do the books say we have?"

Holds	One posting for each warehouse movement, with its status (posted or error)
Calls the item	**ItemNumber**
Does not know	Anything WMS did not send it. A missing posting understates stock and finance
Simulator tab	Inventory postings

> **In the simulator** all four systems run side by side, one tab per record type, so a change in one system and the flag it causes on the worklist can be watched together. Section 7 walks through each tab.

3 · WHERE THE DATA COMES FROM

Five Bimbo reports, three added record types

Bimbo provided five daily reports for one Saturday sales day (12 Sep 2026). The simulator copies their columns exactly, so the agent reads simulated and real data in the same way. To show live warehouse activity we added three record types that the reports do not contain.

SIMULATOR TAB	ONE ROW MEANS	CAME FROM	ROW ID	REAL	SIM
MIPS **Order lines**	One plan line: a customer order or a forecast	BIMBO MIPS Order Data	`ML-0001` added	157	17
ODI **Finished goods**	One item made in one production run	BIMBO ODI Finished Good Data	`FG-0001` added	86	14
ODI **Batches**	The dough mixing for one run code	BIMBO ODI Batch Data	`BT-0001` added	36	6
WMS **Item lines**	One customer's order for one item	BIMBO Order Summary by Item	`WL-0001` added	364	23
WMS **Orders**	One customer's whole order and its progress	BIMBO Fill Rate Report	Order Nbr, real	18	4
WMS **Tickets**	One receipt of product from a production run	ADDED no report exists	`TK-0001`	–	11
WMS **Transactions**	One stock movement: receipt, shipment, return or freezer	ADDED no report exists	`WT-0001`	–	11
ORACLE **Postings**	Oracle's record of one warehouse movement	ADDED no report exists	`OT-0001`	–	11

"Real" is Bimbo's reports for Sat 12 Sep 2026: 80 items, 76 of them in all three of MIPS, ODI and WMS, and 16 customers. "Sim" is the simulator's starting day, Sat 7 Nov 2026, before any scenario runs: 12 items and 4 customers.

What we added, and why

Row IDs on four record types

The MIPS, ODI and WMS item reports have no line numbers, but a live change has to say which row it changes. We added `ML-`, `FG-`, `BT-` and `WL-` IDs. WMS order numbers are real and were kept. The real systems will need permanent IDs for these rows (section 12).

A sales date on every WMS item line

The printed item report shows the sales date once, at the top. Once the report is split into one row per customer line, each row needs its own copy. This is the only column we added to an existing report.

Tickets, transactions and Oracle postings

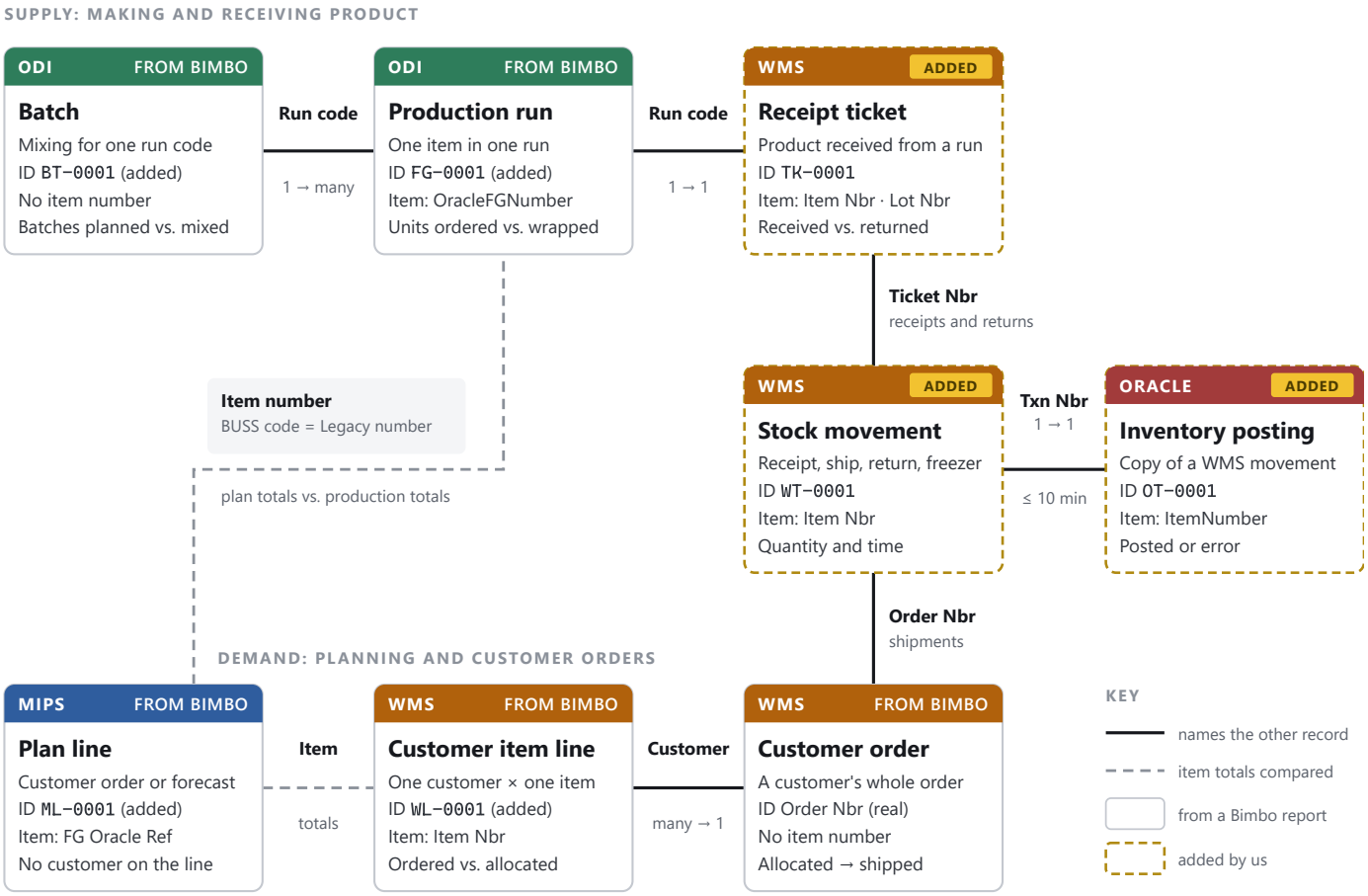
The reports stop at *ordered, allocated, shipped*. To check that every warehouse movement reaches Oracle, and that returns reduce what was received, we needed the movements themselves. **Their shape is our assumption:** no Bimbo file exists to compare them with.

Invented values, small on purpose

12 items, 4 customers (two distribution centers, two store-delivery routes). Ten items each carry one planted problem and two are clean, so each problem is easy to point to. Three more problems are plant-wide: shipping, mixing and data format.

How the eight record types connect

A **solid line** means one record names the other directly, for example a ticket that names its production run. A **dashed line** means the records share only the item number, so the agent adds them up per item and compares the totals.



Reading the map

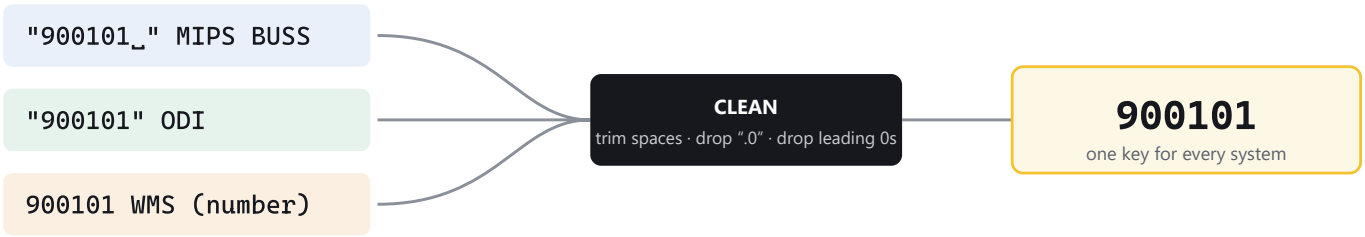
- **Supply chain (top):** each dough batch record covers one run code, and one run code can make several items. Each run that wraps product produces one receipt ticket. Each ticket produces a receipt movement, and every movement should reach Oracle as a posting within 10 minutes.
- **Demand chain (bottom):** each customer has one order header and one item line per item ordered. Shipping an order creates one shipment movement per item, and each movement names the order.
- **The plan line has no solid links.** MIPS does not know customers, tickets or orders. It is compared with the other systems through the item number and the sales day.
- **Customer order to item line uses the customer name.** The item report does not show an order number. In the real report two customers have two orders each, so the name alone cannot say which of the two orders a line belongs to (section 12).

5 · THE COMMON THREAD

One item number, written three ways

The item number is in six of the eight record types, but each system writes it in its own way. Before comparing anything, the agent cleans every item number to one standard form.

SYSTEM · RECORD	COLUMN	HOW THE REAL REPORT WRITES IT	SIMULATED EXAMPLE
MIPS plan line, customer order (BUSS)	FG Oracle Ref	Text, 4–6 digits, padded with spaces to 7 characters	`"900101 "` (trailing space)
MIPS plan line, forecast (OCloud)	FG Oracle Ref	Text, 4–6 digits, no padding	`"900101"`
ODI production run	OracleFGNumber	Text, 4–6 digits	`"900101"`
WMS customer item line	Item Nbr	**A number**, not text	`900101`
WMS ticket, movement	Item Nbr	No real report. Assumed to match the item line	`"900101"`
ORACLE posting	ItemNumber	No real report. Assumed to match Oracle Fusion	`"900101"`



Cleaning the item number. The three spellings become one key. The original MIPS spelling is kept so that check R15 can report the format difference.

Other identifiers that help, and two that do not

✓ BUSS code = Legacy number

The old six-digit code (`009101`) is in MIPS as **FG BUSS Code** and in ODI as **LegacyFGNumber**. In the real data the two are equal for every matched item. The agent keeps it but does not need it to match records.

✓ Run code

OracleTypeCode on a production run, **Oracle Type** on a batch and **Run Code** on a ticket, for example `SU900110`. A code ending in "A" is a run that was moved to another line. One code can cover several items: up to 13 in the real data.

✗ Descriptions

The same bread has a different name in each system. In the real data the MIPS and ODI names differ for 70% of items, and the WMS and MIPS names differ for 66%. The agent never matches on names. The simulator uses one "SIM …" name everywhere to keep the examples readable.

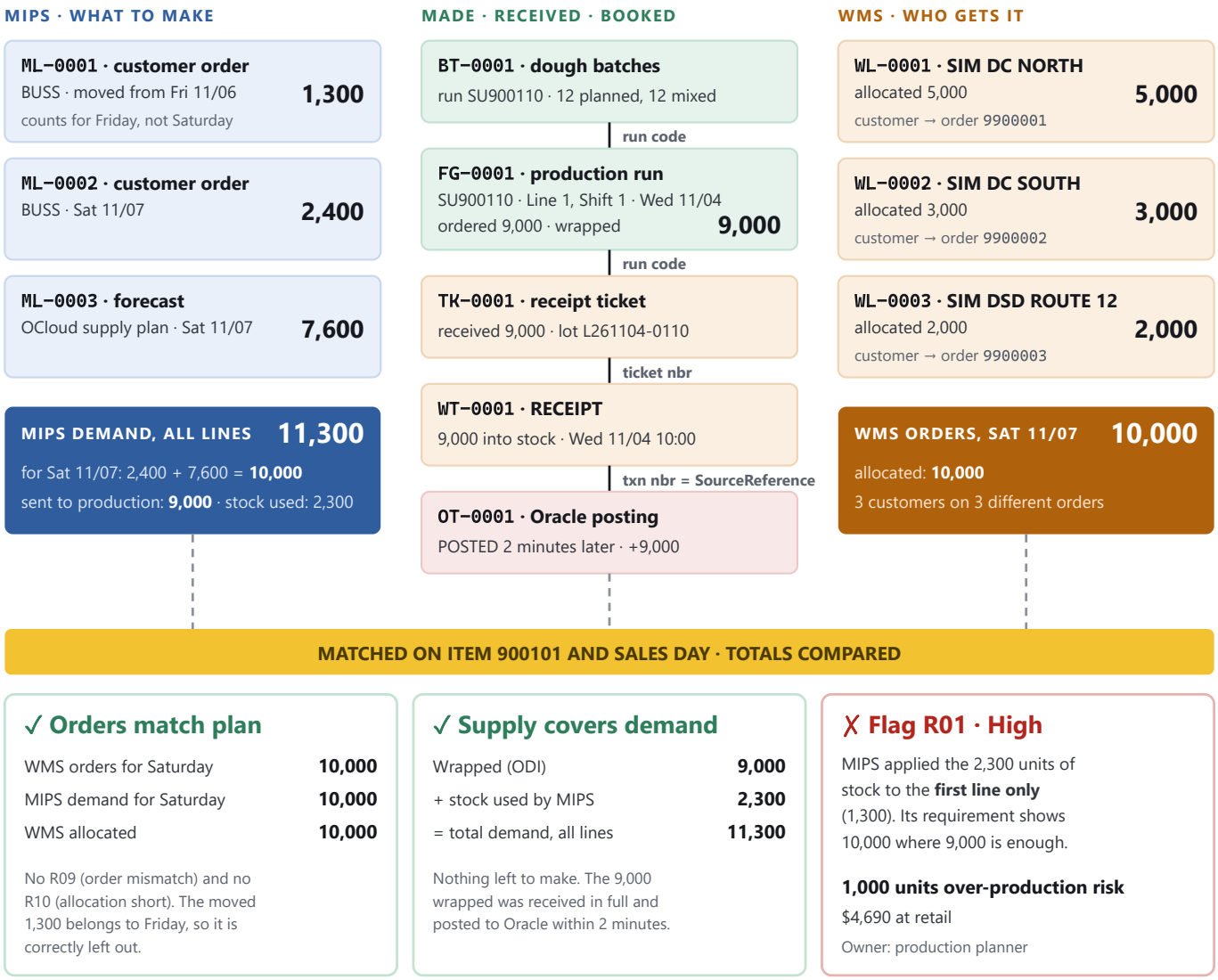
✗ Line numbers

No MIPS line number appears in WMS, and no WMS line number appears in MIPS. That is why the agent compares **item totals for each sales day**, as the next section shows.

6 · ONE ITEM THROUGH EVERY SYSTEM

Following item 900101 line by line

SIM 100% WW BREAD 24Z has records in all eight record types: 3 plan lines, 1 batch, 1 run, 1 ticket, 1 receipt, 1 Oracle posting and 3 customer lines on 3 orders. This page shows how the agent connects them.

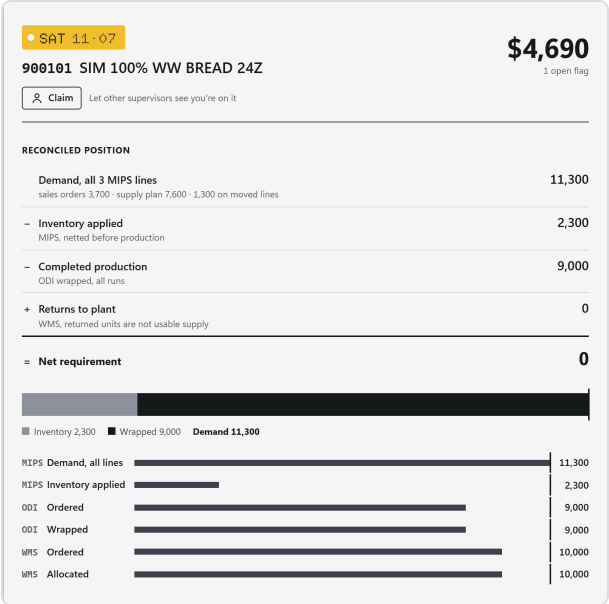


Item 900101 on the simulator's starting day. No single line matches another line directly. The agent uses the solid links within each chain and the item total across the chains.

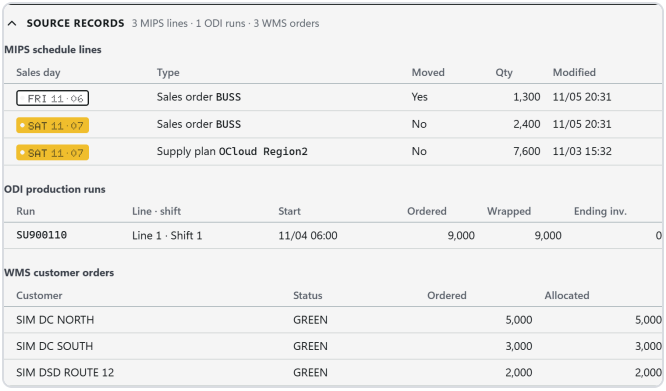
> **Why not pair the lines one to one?** MIPS splits Saturday's 10,000 by *kind of demand*: 2,400 of customer orders and 7,600 of forecast. WMS splits the same 10,000 by *customer*: 5,000, 3,000 and 2,000. The two splits do not line up, and neither system stores the other's line ID. The total for each item and sales day is the only figure the systems share, and for this item those totals agree.

The same item on the worklist

For each item and sales day the worklist shows the result of that matching: demand, stock used, wrapped and returns, then one bar per system. Bars that end at different points show a difference.



Worklist, item 900101, "Reconciled position". Demand 11,300 − stock 2,300 − wrapped 9,000 = nothing more to make. The bars show MIPS, ODI and WMS side by side.



"Source records" at the foot of the same panel. The 3 MIPS lines, the run and the 3 customer lines behind the totals, so a supervisor can check the evidence without opening each system.

> **Sales day matters as much as the item.** The Friday line (white tag) is an order moved from Friday onto Saturday's schedule (**Order Moved** = Yes). Its demand still belongs to Friday, so the agent counts it there. The two Saturday lines make up the 10,000 that WMS has on order for Saturday.

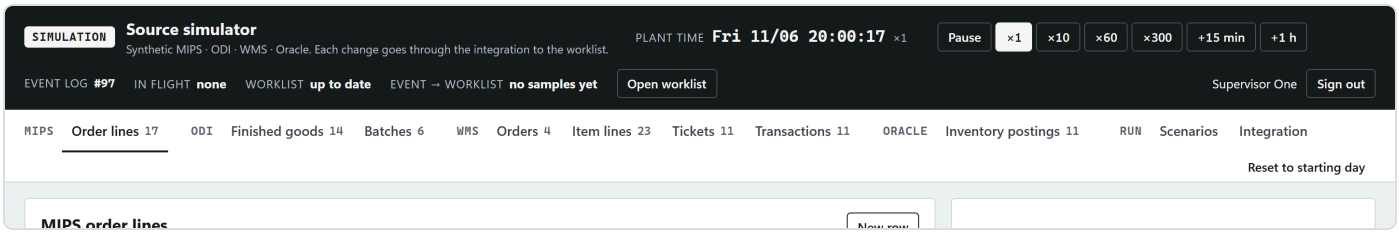
Every record behind item 900101

SYSTEM	RECORD	WHAT IT SAYS	UNITS	LINKED TO
MIPS	ML-0001	Customer order (BUSS), moved from Fri 11/06	1,300	item 900101
MIPS	ML-0002	Customer order (BUSS), Sat 11/07	2,400	item 900101
MIPS	ML-0003	Forecast (OCloud), Sat 11/07	7,600	item 900101
ODI	BT-0001	Dough for run SU900110: 12 batches planned, 12 mixed	–	run code SU900110
ODI	FG-0001	Run SU900110, Line 1, Shift 1: ordered 9,000, wrapped 9,000	9,000	item + run code
WMS	TK-0001	Received from run SU900110, lot L261104-0110	9,000	run code SU900110
WMS	WT-0001	RECEIPT movement	9,000	ticket TK-0001
ORACLE	OT-0001	Posted, 2 minutes after the receipt	+9,000	SourceReference WT-0001
WMS	WL-0001	SIM DC NORTH: ordered and allocated	5,000	order 9900001 (by customer)
WMS	WL-0002	SIM DC SOUTH: ordered and allocated	3,000	order 9900002 (by customer)
WMS	WL-0003	SIM DSD ROUTE 12: ordered and allocated	2,000	order 9900003 (by customer)

Continued in section 9: the SIM DC SOUTH shipment of these 3,000 units never reaches Oracle.

The eight record types in the simulator

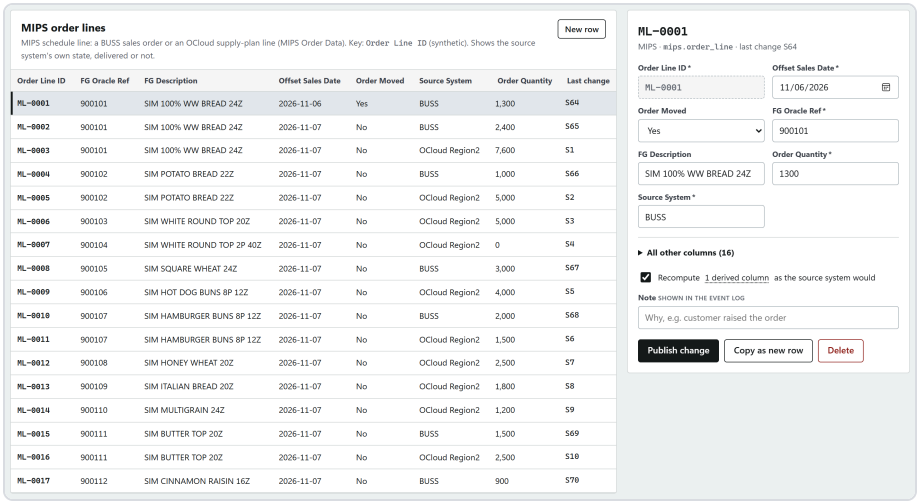
For each record type: what one row means, how it is identified, the fields a warehouse leader should know, and what it links to. In each screenshot the selected row belongs to item 900101.



Source simulator header. Tabs are grouped under MIPS, ODI, WMS and ORACLE, with the row count on each. “Worklist up to date” means the agent has checked every change. On the right of every table, the selected row opens for editing.

MIPS Order lines

BIMBO MIPS Order Data · 157 real, 17 simulated



ONE ROW MEANS

One line of the plan: “for this sales day, make this many of this item”

IDENTIFIED BY

`Order Line ID` ML-0001, added by us. The report has no line number

TWO KINDS OF LINE

BUSS = a customer sales order. **OCloud Region2** = forecast. An item's demand is the sum of all its lines

WATCH FOR

Order Moved = Yes: the line belongs to an earlier sales day. **Units per Container** must agree with WMS (check R11)

LINKS TO

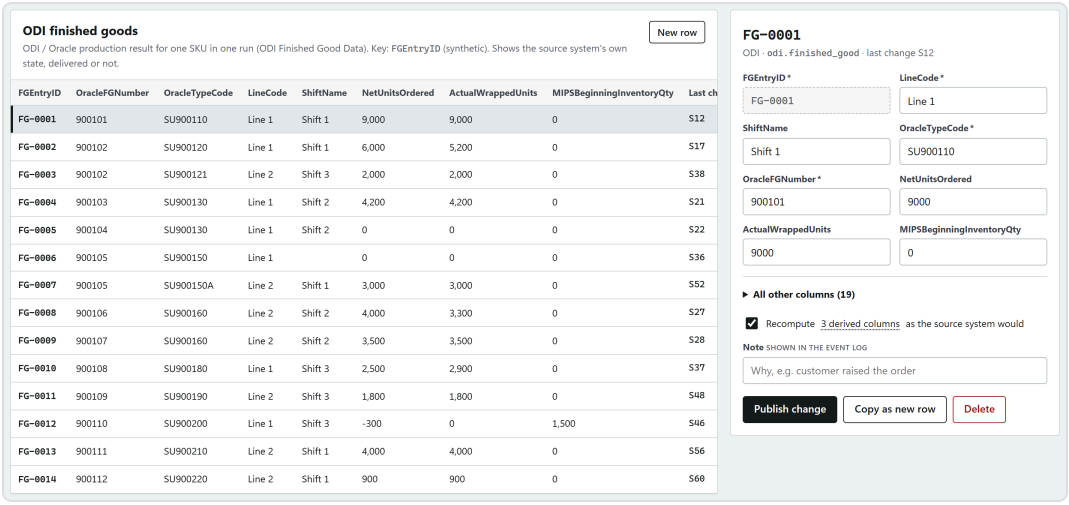
Item number only. There is no customer, order or run on a MIPS line

ALSO CARRIES

Retail price (used for $ at risk), containers per rack, last-modified time

ODI Finished goods (production runs)

BIMBO ODI Finished Good Data · 86 real, 14 simulated



ONE ROW MEANS

One item made in one production run, on one line and shift

IDENTIFIED BY

`FGEntryID` FG-0001, added by us. The run itself is identified by its **run code** (OracleTypeCode)

KEY QUANTITIES

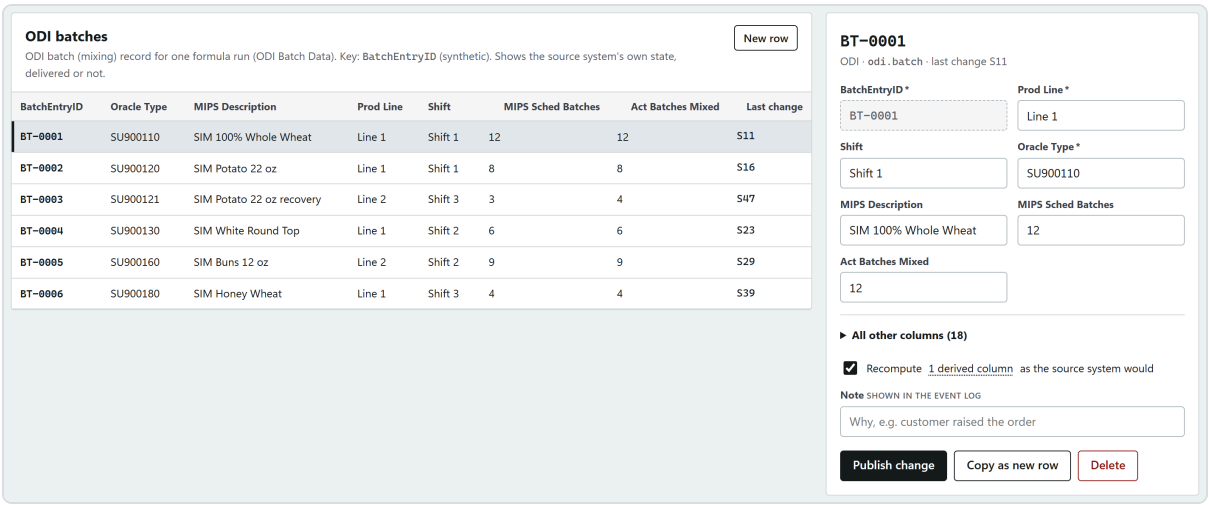
NetUnitsOrdered: what MIPS asked for.
ActualWrappedUnits: what was made. The difference is expected ending stock

WATCH FOR

A zero row with no shift is the ghost of a moved run (R04). A negative order means stock already covers demand (R08)

ODI Batches (dough mixing)

BIMBO ODI Batch Data · 36 real, 6 simulated



ONE ROW MEANS

The dough mixing for one run code: batches planned, batches mixed, pieces expected

IDENTIFIED BY

`BatchEntryID` BT-0001, added by us. There is **no item number**: it links to runs by **Oracle Type** = run code

WATCH FOR

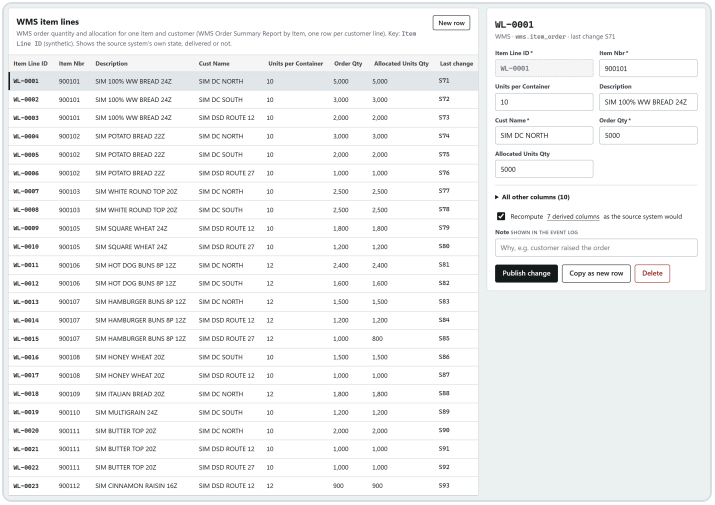
More or fewer batches mixed than planned (R14). In the simulation, BT-0003 mixed 4 batches against 3 planned

LINKS TO

Every production run with the same run code. One code can cover several items

WMS Item lines (customer × item)

BIMBO WMS Order Summary Report by Item · 364 real, 23 simulated



ONE ROW MEANS

One customer's order for one item on the sales day

IDENTIFIED BY

`Item Line ID` WL-0001, added by us. The real report prints one block per item with a subtotal, which we turned into one row per customer

KEY QUANTITIES

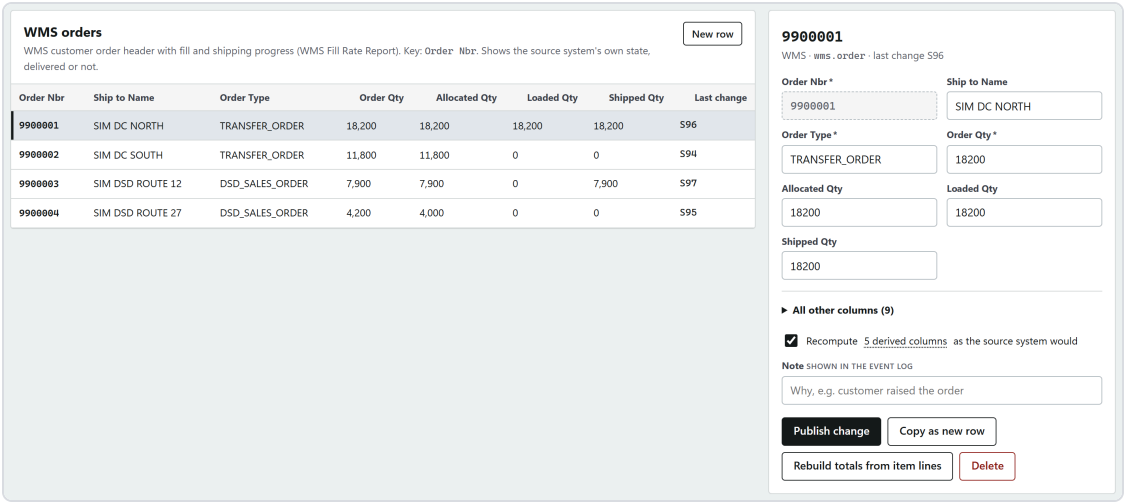
Original Ord Qty, **Order Qty** (after changes), **Allocated Units Qty**, then stacks and trays. Adding up an item's lines gives the WMS total that is compared with MIPS

LINKS TO

The customer's order header, through **Cust Name**. There is no order number on the line

WMS Orders (customer order headers)

BIMBO WMS Fill Rate Report · 18 real, 4 simulated



ONE ROW MEANS

One customer's whole order for the day, and how far it has progressed: allocated → packed → loaded → shipped

IDENTIFIED BY

`Order Nbr`, the real WMS order number. The real report also has a “Total:” row, which the agent keeps separately

ORDER TYPES

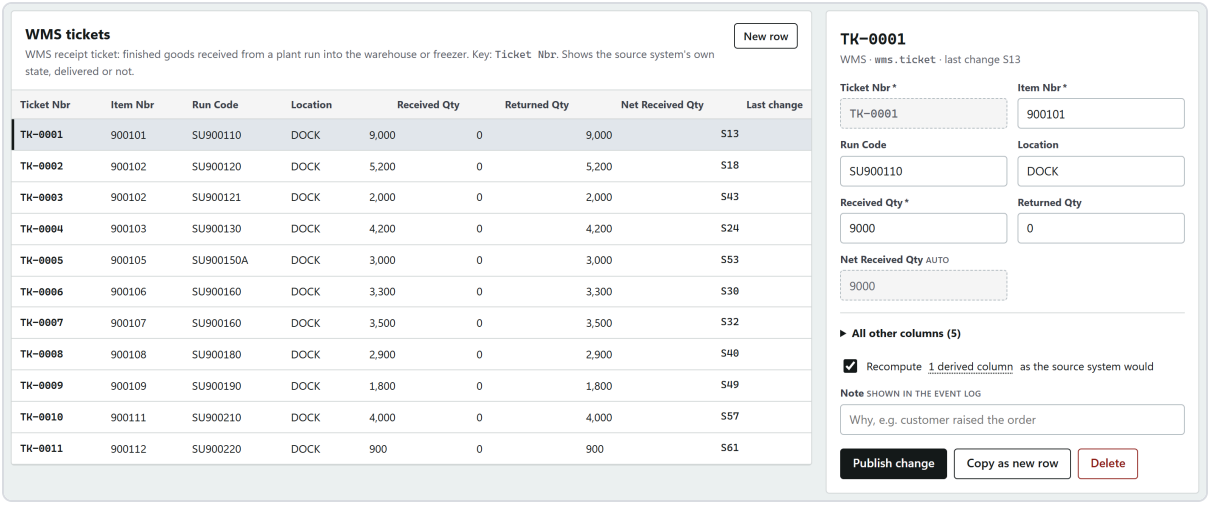
TRANSFER_ORDER to a distribution center, **DSD_SALES_ORDER** for a store-delivery route (with a route number)

WATCH FOR

Shipped with **Loaded Qty** 0 means there was no trailer load scan (R12). Header totals should equal the sum of the customer's item lines (R13)

WMS Tickets (receipts from the plant)

ADDED no Bimbo report · 11 simulated



WMS tickets

WMS receipt ticket: finished goods received from a plant run into the warehouse or freezer. Key: `Ticket Nbr`. Shows the source system's own state, delivered or not.

New row

Ticket Nbr	Item Nbr	Run Code	Location	Received Qty	Returned Qty	Net Received Qty	Last change
TK-0001	900101	SU900110	DOCK	9,000	0	9,000	S13
TK-0002	900102	SU900120	DOCK	5,200	0	5,200	S18
TK-0003	900102	SU900121	DOCK	2,000	0	2,000	S43
TK-0004	900103	SU900130	DOCK	4,200	0	4,200	S24
TK-0005	900105	SU900150A	DOCK	3,000	0	3,000	S53
TK-0006	900106	SU900160	DOCK	3,300	0	3,300	S30
TK-0007	900107	SU900160	DOCK	3,500	0	3,500	S32
TK-0008	900108	SU900180	DOCK	2,900	0	2,900	S40
TK-0009	900109	SU900190	DOCK	1,800	0	1,800	S49
TK-0010	900111	SU900210	DOCK	4,000	0	4,000	S57
TK-0011	900112	SU900220	DOCK	900	0	900	S61

TK-0001

WMS · `wms.ticket` · last change S13

Ticket Nbr * TK-0001 | Item Nbr * 900101

Run Code SU900110 | Location DOCK

Received Qty * 9000 | Returned Qty 0

Net Received Qty AUTO 9000

▸ All other columns (5)

☑ Recompute 1 derived column as the source system would

Note SHOWN IN THE EVENT LOG: Why, e.g. customer raised the order

Publish change | Copy as new row | Delete

ONE ROW MEANS

One delivery of finished product from a production run into the warehouse or freezer

IDENTIFIED BY

`Ticket Nbr` TK-0001. Carries the item, the **run code** and a **lot number**

KEY QUANTITIES

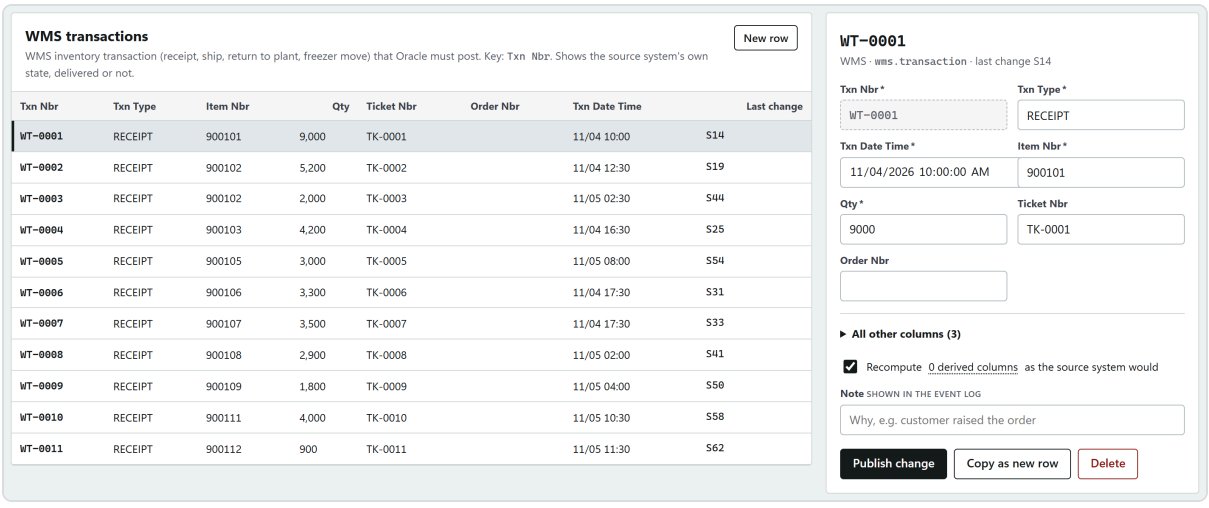
Received Qty − **Returned Qty** = **Net Received Qty**

RULE OF THUMB

One ticket for every run that wrapped product, received 30 minutes after the run ends. Zero runs get no ticket

WMS Transactions (stock movements)

ADDED no Bimbo report · 11 simulated, more as scenarios run



WMS transactions

WMS inventory transaction (receipt, ship, return to plant, freezer move) that Oracle must post. Key: `Txn Nbr`. Shows the source system's own state, delivered or not.

New row

Txn Nbr	Txn Type	Item Nbr	Qty	Ticket Nbr	Order Nbr	Txn Date Time	Last change
WT-0001	RECEIPT	900101	9,000	TK-0001		11/04 10:00	S14
WT-0002	RECEIPT	900102	5,200	TK-0002		11/04 12:30	S19
WT-0003	RECEIPT	900102	2,000	TK-0003		11/05 02:30	S44
WT-0004	RECEIPT	900103	4,200	TK-0004		11/04 16:30	S25
WT-0005	RECEIPT	900105	3,000	TK-0005		11/05 08:00	S54
WT-0006	RECEIPT	900106	3,300	TK-0006		11/04 17:30	S31
WT-0007	RECEIPT	900107	3,500	TK-0007		11/04 17:30	S33
WT-0008	RECEIPT	900108	2,900	TK-0008		11/05 02:00	S41
WT-0009	RECEIPT	900109	1,800	TK-0009		11/05 04:00	S50
WT-0010	RECEIPT	900111	4,000	TK-0010		11/05 10:30	S58
WT-0011	RECEIPT	900112	900	TK-0011		11/05 11:30	S62

WT-0001

WMS · `wms.transaction` · last change S14

Txn Nbr * WT-0001 | Txn Type * RECEIPT

Txn Date Time * 11/04/2026 10:00:00 AM | Item Nbr * 900101

Qty * 9000 | Ticket Nbr TK-0001

Order Nbr

▸ All other columns (3)

☑ Recompute 0 derived columns as the source system would

Note SHOWN IN THE EVENT LOG: Why, e.g. customer raised the order

Publish change | Copy as new row | Delete

ONE ROW MEANS

One stock movement that Oracle must also record

IDENTIFIED BY

`Txn Nbr` WT-0001

FOUR TYPES

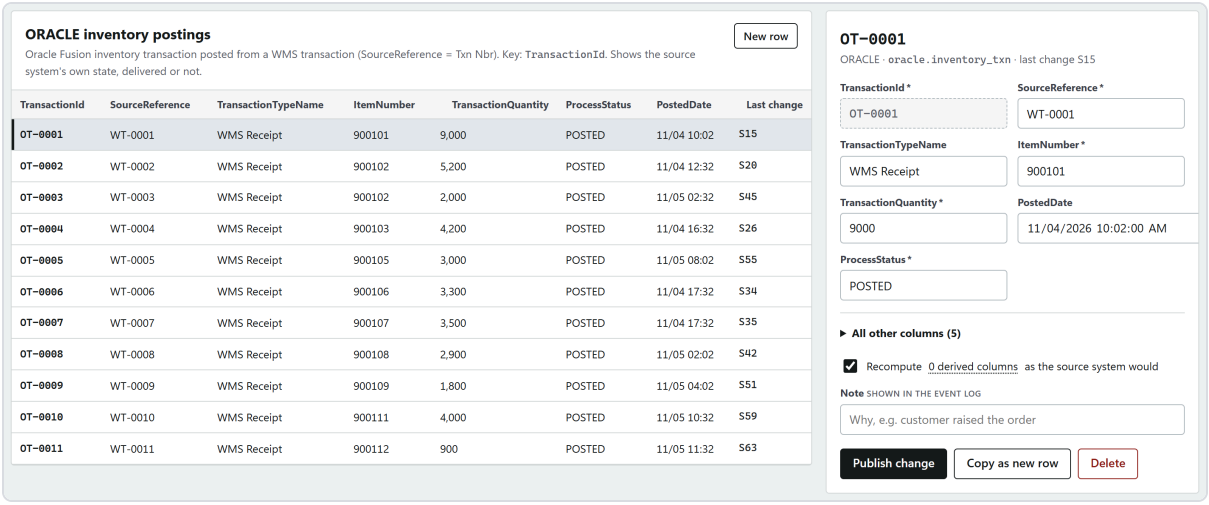
RECEIPT (names a ticket), **SHIP** (names an order), **RETURN_TO_PLANT** (names a ticket), **FREEZER_OUT**

WATCH FOR

Returns that are not taken off the ticket (R19). Freezer stock that MIPS cannot see (R07)

ORACLE Inventory postings

ADDED no Bimbo report · 11 simulated



ONE ROW MEANS

Oracle Fusion's record of one WMS movement

IDENTIFIED BY

`TransactionId` OT-0001. **SourceReference** holds the WMS Txn Nbr it records

KEY FIELDS

TransactionQuantity: negative for shipments and returns.
ProcessStatus: POSTED or ERROR, with an error message

WATCH FOR

A movement with no POSTED row after 10 plant minutes (R18). Until it posts, Oracle stock and finance are wrong

Quick reference: identifiers at a glance

RECORD	ITS OWN ID	ITEM NUMBER COLUMN	OTHER LINKS IT CARRIES
MIPS Plan line	`ML-0001` added	FG Oracle Ref	FG BUSS Code · sales date
ODI Production run	`FG-0001` added	OracleFGNumber	Run code · LegacyFGNumber · line, shift
ODI Batch	`BT-0001` added	none	Run code (Oracle Type)
WMS Customer item line	`WL-0001` added	Item Nbr	Customer name · sales date
WMS Customer order	Order Nbr, real	none	Customer name · route
WMS Ticket	`TK-0001` added	Item Nbr	Run code · lot
WMS Movement	`WT-0001` added	Item Nbr	Ticket Nbr or Order Nbr · lot
ORACLE Posting	`OT-0001` added	ItemNumber	SourceReference = WMS Txn Nbr · lot

8 · HOW A CHANGE TRAVELS

From a change in one system to a flag on the worklist

Today Bimbo sends five Excel reports once a day. In the live setup, each change in MIPS, WMS or Oracle is sent as it happens. The simulator plays the part of the live feed, so the worklist can be tested against changes made during a shift.

1 · Change
A row changes in one system: someone edits it in the simulator, a scenario runs, or a correction is approved

›

2 · Same-row totals
The system recalculates its own fields in that row only, for example Short Qty or Net Received

›

3 · Integration
The interface carries the change. It can be set to delay, lose, duplicate or reorder changes

›

4 · Event log
Every change is kept in order: S-numbers for changes made, #-numbers for changes delivered

›

5 · Worklist
Within about a second it adds up every item again, runs all 19 checks and updates open screens

Nothing passes from one system to another automatically

If WMS raises a customer's order, MIPS keeps its old plan, the WMS order header keeps its old total, and ODI keeps its run. Only three things change other systems: a **scenario** script, a **person** editing in the simulator, or a **correction approved** on the worklist. The plant works the same way, and that is why differences appear.

Corrections flow back only after approval

When a planner or manager approves a fix on the worklist, it goes back through the integration to the system it corrects. The worklist marks it **Confirmed** only when the change has come back through the normal feed, so a correction that never arrived cannot be counted as done.



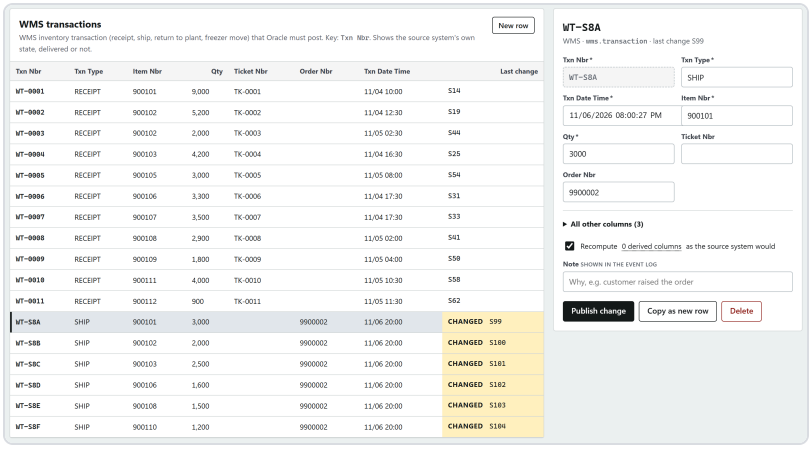
Source changes feed in the simulator. Newest first, across all systems, with what the integration did with each change and whether the worklist has checked it. The entries shown come from scenario S8 (next section).

> **Daily reports and the live feed meet at the same point.** The five Excel files and the live change feed are both converted into the same records, by the same code, using Bimbo's own column names. That is why the simulator copies those column names exactly: everything the agent learns from the simulation applies to the real reports, and the other way round.

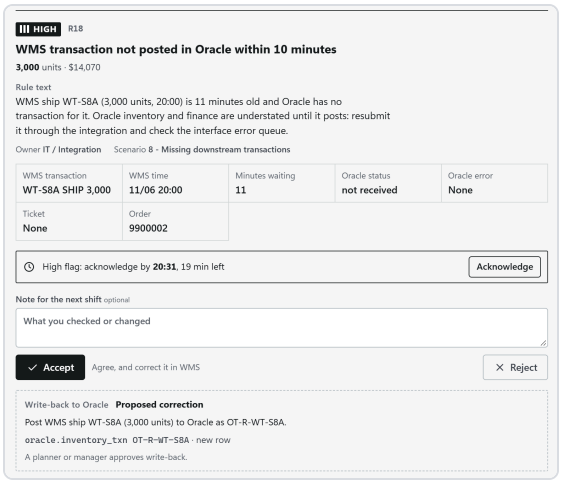
A shipment that never reached Oracle

The SIM DC SOUTH trailer ships order 9900002, which has six items. WMS records one shipment movement per item. Oracle posts four of them, rejects one, and never receives the sixth: item 900101's 3,000 units.

PLANT TIME	WHAT HAPPENS	RECORDS
20:00	WMS ships order 9900002: Loaded and Shipped become 11,800. One SHIP movement per item, each naming the order	`9900002`, `WT-S8A` to `WT-S8F`
20:02	Oracle posts four shipments. It rejects 900110's (“Lot not found in subinventory FG”) and never receives 900101's	`OT-S8B` to `OT-S8E` POSTED, `OT-S8F` ERROR
20:10	10 plant minutes after the shipment, the worklist raises **R18** twice, both High	R18 on 900101 and on 900110



WMS transactions after S8. Six new SHIP rows, each naming order 9900002 (highlighted as changed). The selected row, WT-S8A, is item 900101's 3,000 units.



The R18 flag on item 900101. It names the movement, the order, how long it has waited and Oracle's status. It also proposes the fix: post the shipment to Oracle once a planner or manager approves.

How the agent found it

1. Each WMS movement is matched to Oracle postings by **Txn Nbr = SourceReference**.
2. WT-S8A has no matching posting at all. WT-S8F has one, but it is in ERROR, which counts as not posted.
3. After 10 plant minutes both are overdue, so each raises R18 on its item's row.
4. The $ at risk uses the item's retail price. That is why 900101 moves to the top of the worklist.

> **Why this matters to the warehouse:** the product has left the building, but until Oracle posts the shipment, Oracle still counts it as stock and finance is wrong. Without the agent, this kind of gap is usually found days later, at stock count or month end.

Nineteen checks, grouped by the systems they compare

Every check runs again after every change. A flag names the item, the units and dollars at risk, who owns the fix, and the records it is based on.

CHECK	WHAT IT CATCHES	SEVERITY
MIPS plan compared with ODI production		
R01	Stock was applied to the first order line only, so MIPS asks for more than is needed	High
R02	MIPS took stock off demand, but nothing records where that stock is	Medium
R03	A production line with zero demand and zero output: a “blind” line	Low
R04	A run was moved to another line and left a ghost copy behind	Medium
R05	A recovery run made far more than the shortfall	High / Medium
R06	Surplus was not carried forward to the next sales day	Medium / Low
R08	Stock already exceeds demand, so the run is not needed	Medium
R16	ODI's copy of the MIPS sales-order quantity is out of date	Medium
MIPS plan compared with WMS warehouse		
R07	WMS shipped stock that MIPS cannot see, often freezer stock	High / Medium
R09	MIPS and WMS disagree on the order quantity, often because a screen has not refreshed	Medium
R10	WMS allocated less than was ordered	Medium
R11	MIPS and WMS disagree on units per tray	Medium
Inside WMS		
R12	Orders shipped with no trailer load scan	High
R13	The order headers and the item lines do not add up to the same totals	Medium
ODI mixing		
R14	More or fewer dough batches were mixed than planned	Low
Data quality, all systems		
R15	Item numbers are formatted differently between systems	Low
R17	A number is far outside the plant's normal range, possibly a typing error	Medium
WMS compared with ORACLE (simulation only until the live feed exists)		
R18	A warehouse movement has not been posted in Oracle within 10 minutes	High
R19	A return to the plant was not taken off the received quantity	High / Medium

The worklist ranks flags by **dollars at risk × severity × how reliable the check has proved**. A check's reliability goes down when planners reject its flags. Decisions made in the simulation are kept apart and never affect what is learned from real plant data. High flags must be acknowledged within 15 plant minutes.

How close the simulator is to Bimbo's data

The simulator follows Bimbo's reports exactly in structure: the same columns, the same identifiers and the same quirks that the checks depend on. Its values are invented and its volume is much smaller.

✓ The same as the real reports

- Column names and order in all five reports
- How records are matched: the item number, BUSS code = Legacy number, run codes on runs and batches, customer names
- Moved lines carry the earlier sales date, BUSS item numbers are space-padded, moved runs end in “A” and leave a zero-quantity copy
- Expected ending stock, damaged units, short quantity and the fill percentages are calculated as the plant calculates them

✗ Different, and noticeable on a WMS screen

- **Stacks and trays:** the simulator uses 21 trays per stack for every item. Real WMS uses the item's containers per rack, which is 17 for most 10-unit items
- **Status color:** the simulator turns an order YELLOW when it is short. The real reports show GREEN even on short orders
- **Number lengths:** real store-delivery orders have 14 digits and routes 8. The simulator uses 7 for both
- **Descriptions:** one “SIM …” name everywhere, where the real systems use different names

Volume

MEASURE	REAL DAY (12 SEP 2026)	SIMULATOR START (7 NOV 2026)
Items in total · in all three of MIPS, ODI, WMS	80 · 76	12 · 11
MIPS plan lines (customer orders + forecast)	157 (89 + 68)	17 (7 + 10)
Plan lines per item	1 to 4	1 to 3
ODI production runs · batch records	86 · 36	14 · 6
WMS customer item lines · customer orders · customers	364 · 18 · 16	23 · 4 · 4
Receipt tickets, movements, Oracle postings	no report	11 each
Flags raised	104, from 16 of the checks	14, one planted problem each

What the real reports showed. These observations come from Bimbo's own sample day (counts only):

- **Loaded Qty is 0 on every order** in the Fill Rate report, while Packed, Allocated and Shipped are equal. Either trailer load scans are not captured, or the report does not show them. Check R12 therefore applies to every order.
- **Status is GREEN on every order**, including 5 orders that are short. The color cannot be relied on to show a shortfall.
- **Order headers and item lines disagree for 4 customers**, so the plant totals of the two WMS reports differ (R13).
- Units per container differ between MIPS and WMS for 3 items (R11).

Assumptions to check with Bimbo IT

The live feed depends on these assumptions about the real systems. Each one was needed to build the simulator. Where an assumption proves wrong, the matching in this guide changes with it.

#	WE ASSUMED	WHY IT MATTERS ON THE FLOOR
A1	Each MIPS plan line has a permanent line ID (the simulator uses ML-0001)	Without one, a live change cannot say which line it changes
A2	MIPS keeps an item's lines in a fixed order	R01 depends on which line is “first”
A3	ODI runs and batches have permanent IDs and are updated in place as a run progresses	The same reason as A1, for production
A4	WMS gives order headers and order lines separately, each with its own number. Ideally each line also carries its order number	Today lines link to orders only by customer name, and two real customers have two orders each
A5	Every system records when each row changed, and in what order	Needed to show how up to date the worklist is, and for the 10-minute Oracle check
A6	Column names and meanings match today's reports	The agent reads every source by column name
A7	Each change arrives with the complete row, not only the changed fields	If a change is lost, the next change to the row repairs it
A8	Oracle postings carry the WMS transaction number and a posted or error status	R18 matches warehouse movements to Oracle on this number
A9	WMS tickets keep a returned quantity, and a return names the ticket it reduces	R19 compares the two
A10	The 10-minute posting window is measured from the WMS movement time, in plant time	Decides when R18 raises a flag
A11	Oracle Integration can accept a correction as a row change, and check that the row has not changed in the meantime	Needed to send approved corrections back safely

Questions worth asking the warehouse team

- Are trailer load scans recorded somewhere that the Fill Rate report does not show? (Loaded Qty is 0 on every real order.)
- What turns an order or item line from GREEN to YELLOW in WMS? The real reports never show YELLOW, even on short orders.
- How are receipts, returns to the plant and freezer withdrawals recorded today, and what reference reaches Oracle with each?
- When a customer has two orders on one sales day, how can an item line be tied to the right order?

Terms used in this guide

MIPS	The bakery's planning system: what to make, for which sales day
BUSS	Customer sales orders as they reach MIPS. **FG BUSS Code** is the item's old six-digit code
OCloud	Oracle supply-planning (forecast) lines in MIPS
ODI	Production records: what each run was asked to make, and what it mixed and wrapped
WMS	The warehouse system: customer orders, allocation, tickets, movements and shipments
Oracle Fusion	The company's inventory and finance system. Every WMS movement must post here
Oracle Integration	The interface that carries changes and corrections between systems. The simulator stands in for it
Item / SKU	One finished product, for example 900101. The common thread across all systems
Sales day	The day the product is sold. Plan lines, orders and flags all belong to a sales day
Moved line	A MIPS line that kept today's schedule date but belongs to an earlier sales day
Run code	The formula and run identifier (for example SU900110) shared by runs, batches and tickets. A code ending in "A" is a moved run
Ticket	A WMS receipt of finished product from a production run
Movement / transaction	A stock change in WMS: receipt, shipment, return to plant or freezer withdrawal
Posting	Oracle's record of one WMS movement, with status POSTED or ERROR
Row ID	The ID that identifies one record. Some IDs come from the real systems (Order Nbr). Others were added by us because the reports have none (ML-, FG-, BT-, WL-)
Reconciled position	The worklist's per-item summary: demand − stock used − wrapped + returns = what is still needed or surplus
Flag	One problem found by a check, with the evidence and the recommended action
Source change	A change made in a source system, numbered S1, S2 …
Event	A change as delivered to the worklist, numbered #1, #2 …
Scenario	One of eight scripted stories (S1–S8) that create a known problem so that its flag can be seen appearing
Write-back / correction	A fix sent from the worklist to a source system after a planner or manager approves it
Plant time	The simulator's clock. It can run faster than real time to test the time limits

Related documents in the project: **SYSTEMS_GUIDE.md** (how to use the worklist and the simulator), **SOURCE_DATA_MAPPING.md** (column-by-column comparison with Bimbo's reports), **EVENT_CONTRACT.md** (the technical live-feed contract for IT).